

JOICode Side-by-Side Report · gpt41_mini

Side-by-side GT vs Generated JOICode report for **gpt41_mini**. This report is rendered from row-level benchmark outputs and is print-friendly for PDF export.

ROWS 3	AVG DET 66.2004	DET PASS 0.0%	GT EXACT 0.0%	AVG PROMPT TOKENS 16996.0
AVG COMPLETION TOKENS 94.3	AVG LLM LATENCY 2.138s	P50 LLM LATENCY 2.235s	PEAK VRAM 0.000 GB	OOM COUNT 0

RUN METADATA

Results Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_blocks_equivalence_20260508/retrieval_gpt41_mini_cat7
Output Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_blocks_equivalence_20260508/retrieval_gpt41_mini_cat7/side_by_side_report_gpt41_mini_failures_only
Suite	paper_with_cloud_ref
Genome	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/genomes/example_genome.json
Created At	2026-05-08T14:32:24.771919
Categories	7
Limit/Category	-
Rows	181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230

TOP FAILURE REASONS

- gt_mismatch
- numeric_grounding
- semantic
- dataflow
- gt_receiver_coverage
- gt_service_coverage
- extraneous

MODEL RUNTIME SNAPSHOT

MODEL GPT-4.1-mini	RESOLVED MODEL -	WORKER PYTHON -	SERVICE LIST service_list_ver2.0.1.json	CANDIDATE K 1
REPAIR ATTEMPTS 0	DET PROFILE openai			

ROW DETAILS

Row 187 · Category 7
From 10 PM to midnight every 10 minutes, sound the emergency siren.
밤10시부터 자정까지 10분마다 긴급 사이렌을 울려줘.

GT name: From10PMToMidnightEvery10MinutesSound				
Service list: service_list_ver2.0.1.json				
Output name: SoundEmergencySirenEvery10Minutes10PMToMidnight				
Retrieval cats: Siren, ArmRobot, SoundSensor, Charger, Clock, PresenceVitalSensor, AudioRecorder, Speaker, Button, Oven				
DET 58.8013	PASS No	GT EXACT No	SIMILARITY 0.5936	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.0000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 19852
COMPLETION TOKENS 48	TOTAL TOKENS 19900	LLM LATENCY 1.511s	PIPELINE 2.469s	TOK/S 13167.4717
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallba ck	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
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Siren

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode
raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode
raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

RETRIEVED · RANK 1

ArmRobot 5 services · 2 values · 3 functions VALUES ArmRobot_ArmRobotType raw ArmRobotType · arg - · ret ENUM · Allows for the control of the arm robot ArmRobot_CurrentPosition raw CurrentPosition · arg - · ret STRING · Allows for the control of the arm robot FUNCTIONS ArmRobot_Hello raw Hello · arg - · ret VOID · Allows for the control of the arm robot ArmRobot_SendCommand raw SendCommand · arg ENUM · ret VOID · Allows for the control of the arm robot ArmRobot_SetPosition raw SetPosition · arg ENUM · ret VOID · Allows for the control of the arm robot	RETRIEVED · RANK 2
SoundSensor 1 services · 1 values · 0 functions VALUES SoundSensor_Sound raw Sound · arg - · ret DOUBLE · Sound sensor device for measuring sound levels FUNCTIONS None	RETRIEVED · RANK 3
Charger 4 services · 4 values · 0 functions VALUES Charger_ChargingState raw ChargingState · arg - · ret ENUM · The current status of battery charging Charger_Current raw Current · arg - · ret DOUBLE · The current status of battery charging Charger_Power raw Power · arg - · ret DOUBLE · The current status of battery charging Charger_Voltage raw Voltage · arg - · ret DOUBLE · The current status of battery charging FUNCTIONS None	RETRIEVED · RANK 4
Clock 13 services · 12 values · 1 functions VALUES	RETRIEVED · RANK 5

Clock_Date

raw Date · arg - · ret STRING · Provide current date and time

Clock_Datetime

raw Datetime · arg - · ret STRING · Provide current date and time

Clock_Day

raw Day · arg - · ret INTEGER · Provide current date and time

Clock_Hour

raw Hour · arg - · ret INTEGER · Provide current date and time

Clock_IsHoliday

raw IsHoliday · arg - · ret BOOL · Provide current date and time

Clock_Minute

raw Minute · arg - · ret INTEGER · Provide current date and time

Clock_Month

raw Month · arg - · ret INTEGER · Provide current date and time

Clock_Second

raw Second · arg - · ret INTEGER · Provide current date and time

Clock_Time

raw Time · arg - · ret STRING · Provide current date and time

Clock_Timestamp

raw Timestamp · arg - · ret DOUBLE · Provide current date and time

Clock_Weekday

raw Weekday · arg - · ret ENUM · Provide current date and time

Clock_Year

raw Year · arg - · ret INTEGER · Provide current date and time

FUNCTIONS

Clock_Delay

raw Delay · arg INTEGER | INTEGER | INTEGER · ret VOID · Provide current date and time

PresenceVitalSensor

RETRIEVED · RANK 6

7 services · 7 values · 0 functions

VALUES

PresenceVitalSensor_Awakeness

raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Distance

raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime

raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate

raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex

raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence

raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

AudioRecorder

RETRIEVED · RANK 7

5 services · 2 values · 3 functions

VALUES

AudioRecorder_AudioFile

raw AudioFile · arg - · ret BINARY · Record audio

AudioRecorder_RecordStatus

raw RecordStatus · arg - · ret ENUM · Record audio

FUNCTIONS

AudioRecorder_RecordStart

raw RecordStart · arg - · ret VOID · Record audio

AudioRecorder_RecordStop

raw RecordStop · arg BINARY · ret VOID · Record audio

AudioRecorder_RecordWithDuration

raw RecordWithDuration · arg STRING | DOUBLE · ret BINARY · Record audio

Speaker

RETRIEVED · RANK 8

11 services · 2 values · 9 functions

VALUES

Speaker_PlaybackState

raw PlaybackState · arg - · ret ENUM · Speaker device for audio playback and media control

Speaker_Volume

raw Volume · arg - · ret INTEGER · Speaker device for audio playback and media control

FUNCTIONS

Speaker_FastForward

raw FastForward · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Pause

raw Pause · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Play

raw Play · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Rewind

raw Rewind · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_SetVolume

raw SetVolume · arg INTEGER · ret INTEGER · Speaker device for audio playback and media control

Speaker_Speak

raw Speak · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Stop

raw Stop · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_VolumeDown

raw VolumeDown · arg - · ret INTEGER · Speaker device for audio playback and media control

Speaker_VolumeUp

raw VolumeUp · arg - · ret INTEGER · Speaker device for audio playback and media control

Button

RETRIEVED · RANK 9

1 services · 1 values · 0 functions

VALUES

Button_Button

raw Button · arg - · ret ENUM · A device with one or more buttons

FUNCTIONS

None

Oven

RETRIEVED · RANK 10

4 services · 1 values · 3 functions

VALUES

Oven_OvenMode

raw OvenMode · arg - · ret ENUM · Allows for the control of the oven mode

FUNCTIONS

Oven_AddMoreTime

raw AddMoreTime · arg DOUBLE · ret VOID · Allows for the control of the oven mode

Oven_SetCookingParameters

raw SetCookingParameters · arg ENUM | DOUBLE · ret VOID · Allows for the control of the oven mode

Oven_SetOvenMode

raw SetOvenMode · arg ENUM · ret VOID · Allows for the control of the oven mode

GT JOICODE

cron=0 22 * * * · period=600000

```
if ((#Clock).clock_hour == 0) {  
  
    break  
  
}
```

GENERATED JOICODE

cron=*/10 22-23 * * * · period=0

```
(#Siren).siren_setsirenmode("emergency")
```

```
(#Siren).siren_setsirenmode("emergency")
```

FAILURE REASONS

dataflow

gt_mismatch

gt_receiver_coverage

gt_service_coverage

numeric_grounding

semantic

LINE DIFF

GT	Generated
if ((#Clock).clock_hour == 0) {	(#Siren).siren_setsirenmode("emergency")
break	
}	
(#Siren).siren_setsirenmode("emergency")	

Row 212 · Category 7

Check all door locks in Sector 1 every hour; if any one is open, change all lights in that sector to red.

섹터1의 모든 도어락을 1시간마다 확인해서, 하나라도 열려 있으면 해당 섹터의 모든 조명을 빨간색으로 바꿔줘.

GT name: CheckAllDoorLocksInSector1EveryHourIf				
Service list: service_list_ver2.0.1.json				
Output name: CheckSector1DoorLocksAndSetLightsRed				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.7913	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 0.9167	ENUM 1.0000	PROMPT TOKENS 14777
COMPLETION TOKENS 138	TOTAL TOKENS 14915	LLM LATENCY 2.668s	PIPELINE 2.680s	TOK/S 5591.1681
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 4	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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DoorLock

3 services · 1 values · 2 functions

CONNECTED

VALUES

DoorLock_DoorLockState
raw DoorLockState · arg - · ret ENUM · Allow for the control of a door lock

FUNCTIONS

DoorLock_Lock
raw Lock · arg - · ret VOID · Allow for the control of a door lock

DoorLock_Unlock

raw Unlock · arg - · ret VOID · Allow for the control of a door lock

Light

CONNECTED

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

ColorControl

CONNECTED

2 services · 1 values · 1 functions

VALUES

ColorControl_Color

raw Color · arg - · ret STRING · Allows for control of a color changing device

FUNCTIONS

ColorControl_SetColor

raw SetColor · arg STRING · ret VOID · Allows for control of a color changing device

GT JOICODE

cron="" · period=3600000

```
if (all(#Sector1 #DoorLock).doorlock_doorlockstate ==| "open") {  
    all(#Sector1 #Light).light_movetorgb(255, 0, 0)  
}
```

GENERATED JOICODE

cron="" · period=3600000

```
doorlockStates := [  
    (#Sector1 #DoorLock).doorlock_doorlockstate,  
    (#Sector1 #DoorLock).doorlock_doorlockstate  
]  
anyOpen := false  
for state in doorlockStates {  
    if (state == "open" or state == "opening") {  
        anyOpen = true  
        break  
    }  
}  
if (anyOpen) {  
    all(#Sector1 #Light).light_movetorgb(255, 0, 0)  
}
```

FAILURE REASONS

- dataflow
- extraneous
- gt_mismatch
- numeric_grounding
- semantic

LINE DIFF

GT	Generated
if (all(#Sector1 #DoorLock).doorlock_doorlockstate ==	doorlockStates := [
"open") {	
	(#Sector1 #DoorLock).doorlock_doorlockstate,
all(#Sector1 #Light).light_movetorgb(255, 0, 0)	(#Sector1 #DoorLock).doorlock_doorlockstate
	]
}	anyOpen := false
	for state in doorlockStates {
	if (state == "open" or state == "opening") {
	anyOpen = true
	break
	}
	}
	if (anyOpen) {
	all(#Sector1 #Light).light_movetorgb(255, 0, 0)
	}

Row 218 · Category 7

Every hour from midnight to 5 AM, if at least one door is open, turn all hallway lights to 50%.

자정부터 오전 5시까지 1시간마다 체크해서 문이 하나라도 열려있으면, 복도의 조명을 모두 50%로 켜줘.

GT name: EveryHourFromMidnightTo5AMIfAtLeast				
Service list: service_list_ver2.0.1.json				
Output name: TurnHallwayLightsTo50IfDoorOpenMidnightTo5AM				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.5454	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 1.0000	NUMERIC 0.8333	ENUM 1.0000	PROMPT TOKENS 16359
COMPLETION TOKENS 97	TOTAL TOKENS 16456	LLM LATENCY 2.235s	PIPELINE 2.246s	TOK/S 7362.8635
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 4	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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raw Open · arg - · ret VOID · Allow for the control of a door

Light

CONNECTED

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

LevelControl

CONNECTED

4 services · 3 values · 1 functions

VALUES

LevelControl_CurrentLevel

raw CurrentLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

LevelControl_MaxLevel

raw MaxLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

LevelControl_MinLevel

raw MinLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

FUNCTIONS

LevelControl_MoveToLevel

raw MoveToLevel · arg DOUBLE | DOUBLE · ret VOID · Allows for the control of the level of a device like a light or a dimmer switch

GT JOICODE

cron=0 0-5 * * * · period=0

```
if (all(#Door).door_doorstate ==| "open") {  
  
    all(#Hallway #Light).levelcontrol_movetolevel(50, 0)  
  
}
```

GENERATED JOICODE

cron="" · period=3600000

```
if ((#Clock).clock_hour == 5) {  
    break  
}  
if (all(#Door).door_doorstate == "open" or all(#Door).door_doorstate == "opening") {  
    all(#Hallway #Light).light_movetobrightness(50, 0)  
}
```

FAILURE REASONS

- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- numeric_grounding
- semantic

LINE DIFF

GT	Generated
if (all(#Door).door_doorstate == "open") {	if ((#Clock).clock_hour == 5) {
	break
all(#Hallway #Light).levelcontrol_movetolevel(50, 0)	}
	if (all(#Door).door_doorstate == "open" or all(#Door).door_doorstate == "opening") {
}	all(#Hallway #Light).light_movetobrightness(50, 0)
	}